# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6

Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 126 | Find the breadth first. |
| 2 | **(d)** $3\frac{1}{6}$ | Convert to improper fractions or borrow. |
| 3 | **(c)** intersect at two points | Compare 10 with 8+5=13 and \|8-5\|=3. |
| 4 | **(a)** 18 | Lines of symmetry = $360 \div$ angle. |
| 5 | **(a)** $-13$ | Add step by step. |
| 6 | **(a)** 144 | Area becomes 4 times. |
| 7 | **(c)** 1 | Use the LCM of all denominators. |
| 8 | **(c)** 180 | Triangle angle sum. |
| 9 | **(b)** 300 | Multiply. |
| 10 | **(a)** 8 | Add the opposites. |
| 11 | **(c)** 4800 | Area $\times$ rate. |
| 12 | **(c)** 60 | Divide by 3, multiply by 5. |
| 13 | **(b)** 130 | Linear pair. |
| 14 | **(b)** A square | Only the square among these has 4. |
| 15 | **(a)** 0 | Opposites cancel. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $l + b = 35$, so breadth $= 11$ cm (1); Area $= 264$ cm$^2$ (1). | 1+1 |
| 17 | Convert to improper fractions (1); $3\frac{1}{4}$ (1). | 1+1 |
| 18 | External: $9 + 4 = 13$ cm (1); Internal: 5 cm (1). | 1+1 |
| 19 | Sides $= 360 \div 40 = 9$ (1); Order $= 9$ (1). | 1+1 |
| 20 | Add the first two: 7 (1); Then add (-9): $-2$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Inner lawn: 26 m by 16 m (1); Inner area 416 m$^2$ (1); Path area $= 600 - 416 = 184$ m$^2$ (1). | 1+1+1 |
| 22 | Ploughed =\frac{1}{4}+\frac{1}{3}= $\frac{7}{12}$ (1); Remaining $= 1 - \frac{7}{12}$ (1); $= \frac{5}{12}$ (1). | 1+1+1 |
| 23 | Area of square $= 36$ cm$^2$ (1); One triangle is one quarter: 9 cm$^2$ (1); Angles at centre: four right angles, $360^\circ$ (1). | 1+1+1 |
| 24 | Angle $= 360 \div 5 = 72^\circ$ (1); After 3 turns: $216^\circ$ (1); Turns for full rotation $= 5$ (1). | 1+1+1 |
| | **OR:** Start at 0; after 3 down: $-3$; After 5 up: 2 (1); After 4 down: $-2$ (1); Total floors moved $= 12$ (1). | 1+1+1 |
| 25 | First area $= 80$ cm$^2$ (1); Second area $= 144$ cm$^2$ (1); Larger: second, difference $= 64$ cm$^2$ (1). | 1+1+1 |
| | **OR:** Mixed to improper: $\frac{5}{2}$ (1); Add $\frac{1}{3}$ (1); Subtract $\frac{1}{4}$: result $2\frac{7}{12}$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Hall in cm: 1000 by 800 (1); Tiles along each side: 20 and 16 (1); Number of tiles $= 320$ (1); Cost $=$ ₹12800 (2). | 1+1+1+2 |
| | **OR:** Books: \frac{1}{4}$\times 120 = 30$ (1); Rest $= 90$ (1); Bag: 30 (1); Left $= 60$ (2). | 1+1+1+2 |
| 27 | Draw AB = 4 cm (1); Make right angles at A and B (compass method) (1); Mark AD and BC = 4 cm and join DC (1); Half diagonal OA $= 3$ cm (1); Diagonals meet at $90^\circ$ (1). | 1+1+1+1+1 |
| | **OR:** (a) 5 (1); (b) $72^\circ$ (1); (c) 5 (1); (d) $144^\circ$ is 2 turns so it leaves the pattern unchanged (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 9 | Perimeter of square $=4\times$ side. |
| 2 | **(c)** $\frac{5}{8}$ | Fraction left $=1-$ fraction spent. |
| 3 | **(d)** 5 | Outer radius minus inner radius. |
| 4 | **(a)** 60 | The square fits after every multiple of $90^\circ$. |
| 5 | **(c)** 17 | Highest minus lowest. |
| 6 | **(b)** 100 | Side of square $=$ perimeter $\div 4$. |
| 7 | **(d)** $\frac{3}{8}$ | Take the average of the two fractions. |
| 8 | **(d)** 8 | Side $=$ perimeter $\div 4$. |
| 9 | **(a)** 10 | Add. |
| 10 | **(a)** $-30$ | Subtract the descent. |
| 11 | **(a)** 256 | Inner side $= a - 2 \times 2$. |
| 12 | **(c)** 17 | Whole times denominator plus numerator. |
| 13 | **(a)** 24 | 4 diameters. |
| 14 | **(a)** 5 | Count the letters with a fold line. |
| 15 | **(d)** -7 and 12 | Check each pair. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Area $= 96$ m$^2$ (1); Cost $= 96 \times 75 =$ ₹7200 (1). | 1+1 |
| 17 | Number $+$\frac{1}{2} of number $=$\frac{3}{2} of number (1); Number $= 18 \div$\frac{3}{2}$= 12$ (1). | 1+1 |
| 18 | $\angle AOD = 180 - 40 = 140^\circ$ (1); $\angle OAB = (180 - 40) \div 2 = 70^\circ$ (1). | 1+1 |
| 19 | square: 4 (1); regular octagon: 8; total 12 (1). | 1+1 |
| 20 | $-6 - 9 = -15$ (1); $9 - (-6) = 15$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Between neighbours: 6 cm (1); First to third: 12 cm (1); Total length $= 6 \times 3 = 18$ cm (1). | 1+1+1 |
| 22 | H: order 2, lines 2; A: order 1, lines 1; T: order 1, lines 1 (1); Compare each pair (1); Counting complete (1). | 1+1+1 |
| 23 | After first: 100; After second: $-150$ (1); After deposit: $-30$ (1); Negative means overdrawn (1). | 1+1+1 |
| 24 | Along the length: 5 squares; along the width: 3 squares (1); Squares $= 15$ (1); Left over area $= 250$ cm$^2$ (1). | 1+1+1 |
| | **OR:** Filled $=$\frac{2}{3}$\times 6 = 4$ L (1); More needed $= 2$ L (1); Empty fraction $\frac{1}{3}$ (1). | 1+1+1 |
| 25 | $l + b = 20$, so $b = 6$ cm (1); It is not a square (2). | 1+2 |
| | **OR:** Angle between marks $= 360 \div 12 = 30^\circ$ (1); Lines of symmetry $= 12$ for a regular arrangement (1); Turn to mark 5: $150^\circ$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | After first jump: 5 (1); After second: 0 (1); After third: $-9$ (1); Distance from 0 $= 9$ (2). | 1+1+1+2 |
| | **OR:** Area of big square $= 400$ cm$^2$ (1); Area of four small squares $= 4 \times 9 = 36$ cm$^2$ (1); Remaining area $= 364$ cm$^2$ (1); Perimeter stays the same as the original $= 4 \times 20 = 80$ cm (1); Cutting corner squares does not change the perimeter (1). | 1+1+1+1+1 |
| 27 | Fraction added $=$\frac{3}{4}$-$\frac{1}{4}$= \frac{1}{2}$ (1); This part is 20 litres (2); Capacity $= 20 \div \frac{1}{2} = 40$ litres (2). | 1+2+2 |

| | | |
|---|---|---|
| | **OR:** Width $= 8 - 5 = 3$ m (1); Outer diameter $= 16$ m (1); Paths across $= 16 \div 3 = 5$ (whole part) (1); Check with the compass-drawn figure (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 9 | Height $= 2\times$ area $\div$ base. |
| 2 | **(d)** $\frac{3}{7}$ | Divide numerator and denominator by their HCF. |
| 3 | **(d)** 7 | Half of AB. |
| 4 | **(c)** 45 | $360\div$ spokes. |
| 5 | **(d)** 4 | Count the positive integers up to b-1. |
| 6 | **(c)** 42 | $2(l+b)$. |
| 7 | **(c)** 21 | Divide, then multiply. |
| 8 | **(a)** touch externally | Compare 13 with 9+4=13 and \|9-4\|=5. |
| 9 | **(a)** 20 | Lines of symmetry = $360\div$ angle. |
| 10 | **(d)** $-3$ | Add step by step. |
| 11 | **(b)** 154 | Area $= limesb$. |
| 12 | **(d)** $2\frac{11}{12}$ | Convert to improper fractions or borrow. |
| 13 | **(a)** 34 | Opposite sides are equal. |
| 14 | **(d)** 225 | Multiply. |
| 15 | **(d)** $-10$ | Add the opposites. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | New side 12 cm; New area 144 cm$^2$; old area 100 cm$^2$ (1); Increase $=$ 44 cm$^2$ (1). | 1+1 |
| 17 | Common denominator 24 (1); $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{8}$ (1). | 1+1 |
| 18 | Rectangle $=$ 70 cm$^2$; Square $=$ 64 cm$^2$ (1); Larger: rectangle (1). | 1+1 |
| 19 | Reflect each x-coordinate: $x \to -x$ (1); New x values: -1, -3, -2 (1). | 1+1 |
| 20 | Integers: -4, -3, -2, -1, 0, 1 (1); Number $=$ 6 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $6-(-4)=10$ (1); $10+(-9)=1$ (1); $1-5=-4$ (1). | 1+1+1 |
| 22 | Inner lawn: 34 m by 19 m (1); Inner area 646 m$^2$ (1); Path area $= 1000-646=354$ m$^2$ (1). | 1+1+1 |
| 23 | Ploughed $=$\frac{2}{5}+\frac{1}{4}$=$ $\frac{13}{20}$ (1); Remaining $= 1-\frac{13}{20}$ (1); $=\frac{7}{20}$ (1). | 1+1+1 |
| 24 | Area of square $=$ 64 cm$^2$ (1); One triangle is one quarter: 16 cm$^2$ (1); Angles at centre: four right angles, 360° (1). | 1+1+1 |
| | **OR:** Angle $= 360\div 6 = 60°$ (1); After 4 turns: 240° (1); Turns for full rotation $=$ 6 (1). | 1+1+1 |
| 25 | Start at 0; after 6 down: $-6$; After 2 up: $-4$ (1); After 7 down: $-11$ (1); Total floors moved $=$ 15 (1). | 1+1+1 |
| | **OR:** First area $=$ 108 cm$^2$ (1); Second area $=$ 140 cm$^2$ (1); Larger: second, difference $=$ 32 cm$^2$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | square: 90° (1); regular octagon: 45° (1); Difference $=$ 45° (1); Yes, the larger count is a multiple of the smaller (2). | 1+1+1+2 |
| | **OR:** Coldest to hottest: -8, -5, -2, 3, 11 (1); Hottest 11, coldest $-8$ (1); Difference $=$ 19 (1); Sum $=-1$ (2). | 1+1+1+2 |
| 27 | Hall in cm: 1200 by 900 (1); Tiles along each side: 30 and 22 (1); Number of tiles $=$ 660 (1); Cost $=$ ₹16500 (2). | 1+1+1+2 |
| | **OR:** Books: \frac{1}{6}$\times 240 = 40$ (1); Rest $=$ 200 (1); Bag: 80 (1); Left $=$ 120 (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.